STPM Physics · Semester 3 · Chapter 21

Sound Waves

21.1 Propagation of Sound Waves

Definitions & laws

- **Sound waves** are longitudinal mechanical waves in which the particles of the medium oscillate parallel to the direction of propagation, forming alternate regions of compression and rarefaction.
- A sound wave needs a material medium; it cannot travel through a vacuum, and there is no net flow of the medium from source to listener.
- The **speed of sound** depends on the elasticity and density of the medium: $v = \sqrt{\frac{B}{\rho}}$ in a fluid and $v = \sqrt{\frac{E}{\rho}}$ in a solid, so v is greater in solids than in liquids and gases.
- In an ideal gas $v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{\gamma p}{\rho}}$, so v is proportional to $\sqrt{T}$ (T in kelvin) and independent of pressure at constant temperature.
- **Intensity** is the rate of energy flow per unit area perpendicular to the direction of propagation; $I = \frac{P}{A}$ and $I \propto (\text{amplitude})^2$.
- Displacement nodes are pressure antinodes and displacement antinodes are pressure nodes; the pressure variation is $\frac{\pi}{2}$ out of phase with the displacement.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Wave speed	$v = f\lambda$	v = wave speed (m s^{-1}); f = frequency (Hz); λ = wavelength (m)
Sound wave equation	$y = A \sin(\omega t - kx)$	$\omega = 2\pi f$ (rad s^{-1}); $k = \frac{2\pi}{\lambda}$ (rad m^{-1}); wave speed $v = \frac{\omega}{k}$ (m s^{-1}); A = amplitude
Speed of sound and temperature	$\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$	T = absolute temperature (K); v = speed of sound (m s^{-1}); $v \approx 331$ m s^{-1} at 0 °C
Intensity of a point source	$I = \frac{P}{A} = \frac{P}{4\pi r^2}; \frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}$	Point source radiating uniformly; I = intensity (W m^{-2}); P = power (W); A = area (m^2); r = distance from the source (m)
Echo / sonar distance	$d = \frac{vt}{2}$	d = distance to the reflector (m); v = speed of sound (m s^{-1}); t = total time for the pulse to travel to the reflector and back (s)

How to answer (MPM style)

- State that sound is a longitudinal wave and name compressions and rarefactions; mention the direction of particle vibration relative to the direction of travel.
- For $y = A\sin(\omega t - kx)$, read off ω and k by comparison, then calculate $v = \frac{\omega}{k}$ and $f = \frac{\omega}{2\pi}$; include units.
- To explain why v differs between media, state that v depends on (elastic modulus)/(density), not on density alone.
- For temperature problems, convert °C to kelvin before taking the ratio of square roots; state the answer to 3 s.f.

Section A traps

Trap: Using Celsius temperatures in $v \propto \sqrt{T}$. The ratio must be of absolute temperatures, e.g. $\sqrt{\frac{303}{273}}$, not $\sqrt{\frac{30}{0}}$.

Trap: Using $v = \frac{\omega}{k}$ with k replaced by λ, or $f = \omega$ instead of $f = \frac{\omega}{2\pi}$; this gives 300 Hz instead of 47.7 Hz.

Trap: Saying a denser medium always carries sound faster. Steel is denser than air but has a far larger elastic modulus, so v is greater.

Trap: Forgetting the factor ½ in echo and sonar problems: the pulse travels to the reflector and back.

Tricks

Trick: Speed of sound ratio: a 30 K rise near room temperature changes v by only about 5%, so any option far from 331×1.05 can be rejected at once.

Trick: In $y = A\sin(\omega t - kx)$, $v = \frac{\text{coefficient of } t}{\text{coefficient of } x}$; no need to find f first.

21.2 Sources of Sound

Definitions & laws

- **Standing (stationary) waves** in air columns and strings are formed by the superposition of a progressive wave and its reflection, giving fixed nodes and antinodes.
- **Resonance** occurs when the driving frequency equals a natural frequency of the system, producing a large-amplitude oscillation.
- In an **open pipe** there is a displacement antinode at each end, so all harmonics are present: $f_n = \frac{nv}{2L}$.

- In a **closed pipe** there is a displacement node at the closed end and an antinode at the open end, so only odd harmonics are present: $f_n = \frac{(2n - 1)v}{4L}$.
- In a stretched string fixed at both ends there is a node at each end; the wave speed on the string is $v = \sqrt{\frac{T}{\mu}}$ and $f_n = \frac{nv}{2L}$.
- The antinode at an open end lies slightly beyond the pipe; this **end correction** makes the effective length of the air column larger than the physical length.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Open pipe harmonics	$f_n = \frac{nv}{2L}$; $\lambda_n = \frac{2L}{n}$	f_n = frequency of nth harmonic (Hz); n = 1, 2, 3, ...; v = speed of sound (m s^{-1}); L = length of pipe (m); λ_n = wavelength (m); pipe open at both ends
Closed pipe harmonics	$f = \frac{v}{4L}, \frac{3v}{4L}, \frac{5v}{4L}, \ldots$; $\lambda_1 = 4L$	f = frequency (Hz); v = speed of sound (m s^{-1}); L = length of pipe (m); λ_1 = fundamental wavelength (m); pipe closed at one end
Stretched string harmonics	$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$	f_n = frequency (Hz); n = 1, 2, 3, ...; L = length of string (m); T = tension (N); μ = mass per unit length (kg m^{-1})
Closed pipe fundamental	$v = 4fL$; $L + e = \frac{\lambda}{4}$	v = speed of sound (m s^{-1}); f = frequency (Hz); L = length of air column at first resonance (m); e = end correction (m); λ = wavelength (m)

- Speed of sound in air ≈ 340 m s^{-1} unless otherwise stated.

How to answer (MPM style)

- Draw the displacement pattern (node/antinode) for the pipe first, then write L in terms of λ: open $L = \frac{n\lambda}{2}$, closed $L = \frac{(2n - 1)\lambda}{4}$.
- Identify the harmonic number by dividing the given frequency by the fundamental; for a closed pipe it must be an odd integer.
- State formula, substitute with L in metres, and give the frequency in Hz to 3 s.f.
- For 'second lowest resonance' of a closed pipe, the answer is $3f_1$, not $2f_1$.

Section A traps

Trap: Applying $f = \frac{nv}{2L}$ to a closed pipe, or $f = \frac{v}{4L}$ to an open pipe. The fundamental of a closed pipe is half that of an open pipe of the same length.

Trap: Counting the 'second harmonic' of a closed pipe as $2f_1$. Even multiples are absent; the next resonance is $3f_1$.

Trap: Using the diameter or the pipe length in cm without conversion: L must be in metres.

Trap: Treating the first overtone and the second harmonic as different. For an open pipe the first overtone is the second harmonic ($2f_1$).

Tricks

Trick: Ratio method: closed fundamental = ½ × open fundamental for equal L and v.

Trick: Frequency ratio test: if $\frac{f}{f_1}$ is an even integer or non-integer, the closed-pipe harmonic is impossible.

21.3 Intensity Level of Sound

Definitions & laws

- **Intensity level** (β) is a logarithmic measure of the intensity of a sound relative to the threshold of hearing: $\beta = 10 \log_{10} \frac{I}{I_0}$, measured in decibels (dB).
- The **threshold of hearing** $I_0 = 1.0\times10^{-12}$ W m^{-2} is the minimum intensity audible at about 1 kHz.
- A difference of 10 dB corresponds to a ratio of intensities of 10; a difference of about 3 dB corresponds to a ratio of 2.
- Intensity levels in dB cannot be added directly; the intensities of incoherent sources add, and the total is then converted to dB.
- For a point source the intensity obeys the inverse square law, $I \propto \frac{1}{r^2}$, because the same power spreads over a sphere of area $4\pi r^2$.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Intensity level	$\beta = 10 \log_{10} \frac{I}{I_0}$	β = intensity level (dB); I = intensity (W m^{-2}); I_0 = 1.0×10^{-12} W m^{-2}
Intensity from intensity level	$I = I_0 \times 10^{\beta/10}$	I = intensity (W m^{-2}); $I_0 = 1.0\times10^{-12}$ W m^{-2}; β = intensity level (dB)
Difference in intensity level	$\beta_2 - \beta_1 = 10 \log_{10} \frac{I_2}{I_1}$	β_1, β_2 = intensity levels (dB); I_1, I_2 = intensities (W m^{-2})
N identical incoherent sources	$\beta = \beta_1 + 10 \log_{10} N$	β = intensity level of N sources together (dB); β_1 = intensity level of one source (dB); N = number of identical incoherent sources

Quantity / law	Formula	Symbols, units & conditions
Level difference with distance	$\beta_1 - \beta_2 = 20 \log_{10} \frac{r_2}{r_1}$	β_1, β_2 = intensity levels (dB) at distances r_1, r_2 (m) from the source; for a point source in free space
Intensity from a point source	$I = \frac{P}{4\pi r^2}$	I = intensity (W m^{-2}); P = power of the source (W); r = distance from the source (m)

How to answer (MPM style)

- Write the formula first, substitute I and I_0 with the same units, evaluate the logarithm, and give β in dB to 3 s.f. (or 2 s.f. if the data are 2 s.f.).
- To find I from β, rearrange: $I = I_0 \times 10^{\beta/10}$; show the power of ten explicitly.
- For several sources, add intensities (W m^{-2}) first, then convert; do not add decibels.
- Show the intermediate ratio (e.g. $\frac{I}{I_0} = 2.5 \times 10^7$) so that the logarithm step is clear.

Section A traps

Trap: Adding decibels: two 60 dB sources do not give 120 dB; they give about 63 dB.

Trap: Dropping the factor 10 (or using 20) in $\beta = 10 \log \frac{I}{I_0}$; 20 log is for amplitude or pressure ratios only.

Trap: Using the natural logarithm ln instead of $\log_{10}$.

Trap: Treating 10 dB as doubling the intensity. 10 dB is ×10; ×2 is about 3 dB.

Tricks

Trick: Memorise: ×2 → +3 dB, ×4 → +6 dB, ×10 → +10 dB, ×100 → +20 dB.

Trick: Doubling the distance from a point source lowers the level by 20 log 2 ≈ 6 dB.

21.4 Beat

Definitions & laws

- **Beats** are the periodic variations in loudness heard when two sound waves of slightly different frequencies are superposed.
- Beats are produced because the two waves alternate between being in phase (constructive interference, loud) and in antiphase (destructive interference, soft).
- The **beat frequency** is the number of loud (or soft) sounds heard per second and equals the magnitude of the difference between the two frequencies.

- The frequency of the resultant sound is the mean of the two frequencies, and its amplitude varies at the beat frequency.
- Beats can be used to compare an unknown frequency with a standard; loading or filing one source changes the beat frequency in a predictable way.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Beat frequency	$f_{beat} = \lvert f_1 - f_2 \rvert$	f_{beat} = beat frequency (Hz); f_1, f_2 = frequencies of the two sources (Hz)
Time between loud beats	$T_{beat} = \frac{1}{f_{beat}}$	T_{beat} = time between successive loud beats (s); f_{beat} = beat frequency (Hz)
Resultant frequency	$f_{av} = \frac{f_1 + f_2}{2}$	f_{av} = resultant frequency (Hz); f_1, f_2 = frequencies of the two sources (Hz)
Resultant displacement	$y = 2A \cos(\pi(f_1 - f_2)t) \sin(\pi(f_1 + f_2)t)$	y = resultant displacement; A = amplitude of each wave; f_1, f_2 = frequencies (Hz); t = time (s); for equal amplitudes A

How to answer (MPM style)

- Define beats and state $f_{beat} = \lvert f_1 - f_2 \rvert$; for explanation questions, mention the changing phase difference and alternate constructive and destructive interference.
- A frequency unknown from a beat frequency has two possibilities, $f \pm f_{beat}$; give both unless extra information is given.
- For the extra information (filing, waxing or loading a fork), state how the beat frequency changed and deduce which option is eliminated.
- For the time interval, use $T = \frac{1}{f_{beat}}$ with the unit in seconds.

Section A traps

Trap: Giving the sum $f_1 + f_2$ or the mean as the beat frequency.

Trap: Giving only one value of the unknown frequency. The beat frequency does not show which frequency is higher.

Trap: Reversing the effect of filing or loading. Filing raises a fork's frequency; loading with wax lowers it.

Trap: Confusing the beat frequency with the interval between beats: 4 Hz means an interval of 0.25 s, not 4 s.

Tricks

Trick: Filing a fork (f increases): if the beat frequency rises, the fork was already higher than the standard; if it falls, the fork was lower. Opposite for waxing.

Trick: Beats need frequencies within a few hertz of each other; a difference of 442 Hz is never a beat frequency.

21.5 Doppler Effect

Definitions & laws

- The **Doppler effect** is the change in the observed frequency of a wave due to relative motion between the source and the observer.
- When a source moves towards a stationary observer, the wavelength is shortened and the observed frequency is higher; when it moves away, the wavelength is lengthened and the frequency is lower.
- When an observer moves towards a stationary source, more wavefronts are intercepted per second, so the observed frequency is higher; moving away gives a lower frequency.
- Speeds are taken relative to the air (medium); the speed of sound in air is unchanged by the motion of source or observer.
- The Doppler shift of waves reflected from a moving object is used in radar speed guns and in medical blood-flow measurement.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Moving source, stationary observer	$f' = \frac{fv}{v \mp v_s}$	f' = observed frequency (Hz); f = source frequency (Hz); v = speed of sound (m s^{-1}); v_s = speed of source (m s^{-1}); upper sign (−) when the source approaches, lower sign (+) when it recedes
Moving observer, stationary source	$f' = \frac{f(v \pm v_o)}{v}$	f' = observed frequency (Hz); f = source frequency (Hz); v = speed of sound (m s^{-1}); v_o = speed of observer (m s^{-1}); plus when the observer approaches, minus when moving away
General Doppler formula	$f' = \frac{f(v \pm v_o)}{v \mp v_s}$	f' = observed frequency (Hz); f = source frequency (Hz); v = speed of sound (m s^{-1}); v_o = speed of observer (m s^{-1}); v_s = speed of source (m s^{-1})
Small-speed approximation	$\frac{\Delta f}{f} \approx \frac{v_{rel}}{v}$	Δf = change in frequency (Hz); f = frequency (Hz); v_{rel} = relative speed of source and observer (m s^{-1}); v = speed of sound (m s^{-1}); for small speeds

- Reflection from a moving surface: apply the formula twice, first with the surface as the observer, then as the source.

How to answer (MPM style)

- Identify who is moving, then choose the formula; write the sign choice explicitly (approach: frequency higher).
- Substitute with all speeds in m s^{-1} and compute f′ to 3 s.f.; state whether it is higher or lower than f.
- Check the sign of the shift against the physical situation before giving the answer.
- For explanation questions, refer to the number of wavefronts reaching the observer per second and the change in wavelength.

Section A traps

Trap: Using $v + v_s$ in the denominator for an approaching source. Approach means $v - v_s$, giving a larger f′.

Trap: Placing the observer's speed in the denominator or the source's in the numerator (swapped roles).

Trap: Using the same frequency formula for both a moving source and a moving observer, which are different when speeds are comparable to v.

Trap: Giving f′ lower than f for an approaching source or observer.

Tricks

Trick: Direction check: approaching → f′ > f. Eliminate any option on the wrong side of f.

Trick: Source: v_s subtracts from the denominator when approaching. Observer: v_o adds to the numerator when approaching.